MACHINE LEARNING PROJECT REPORT

Phishing Link Detector

Classifying websites as phishing or legitimate with a Random Forest

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Project ID	P33
Project title	Phishing Link Detector

At a glance

10,000 websites | 49 candidate features, 15 selected | Random Forest (final: 200 trees) | **97.75% accuracy, 98.00% phishing recall, ROC-AUC 0.9959** on a held-out 20% test set (2,000 sites).

2. Problem Statement

Phishing websites imitate trusted sites to steal passwords and money, and blacklists only catch sites that were already reported. This project builds a Random Forest classifier that predicts whether a website is phishing (1) or legitimate (0) from features of its URL and page. A missed phishing site is the costly mistake, so Recall for phishing is the key metric.

3. Dataset Description and Data Dictionary

`Phishing_Legitimate_full.csv`: 10,000 rows x 50 columns, one website per row, balanced at 5,000 phishing and 5,000 legitimate. All columns are numeric and there are no missing values. The file has 48 features, an id column and the target CLASS_LABEL.

Column group	Examples	Type	Meaning	Use
URL counts and lengths (16)	NumDots, UrlLength, NumDash, PathLevel, HostnameLength	Number	Counts and lengths of URL parts	Keep
URL red flags (10)	AtSymbol, NoHttps, IpAddress, RandomString	0 / 1	1 if the URL has the suspicious pattern	Keep
Link ratios (3)	PctExtHyperlinks, PctNullSelfRedirectHyperlinks	0 to 1	Share of links to other domains or to nowhere	Keep
Page behaviour (13)	InsecureForms, SubmitInfoToEmail, IframeOrFrame	0 / 1	1 if the page shows the behaviour	Keep
Coded RT features (6)	UrlLengthRT, PctExtNullSelfRedirectHyperlinksRT	-1 / 0 / 1	Rule-coded versions of related features	Keep
id	id	Number	Row number	Drop
CLASS_LABEL	CLASS_LABEL	0 / 1	Target: 0 = Legitimate, 1 = Phishing	Target

Table 1. Data dictionary (grouped by column type). The 15 columns actually used are listed in Table 2. Features are pre-extracted from the URL and page, so the model never reads the raw web address.

Dataset at a glance

Item	Value
Rows (websites)	10,000
Phishing / legitimate	5,000 / 5,000
Missing values	0
Duplicate rows	0 with id present; 419 once id is dropped (see Limits)
Unusable column	HttpsInHostname (same value in every row)

Dataset size and quality checks (details in section 4).

4. Data Cleaning and EDA

- **Cleaning:** no missing values and no duplicate rows were found; the `id` column was dropped (identifier only). 419 rows do repeat once `id` is removed (see Limits).
- **New feature:** DomainComplexity = NumDots + SubdomainLevel. **Selection:** `SelectKBest(f_classif)` kept the 15 best of 49 features.

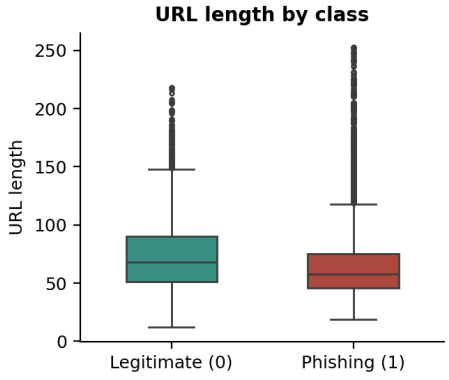


Fig 1. URL length by class.

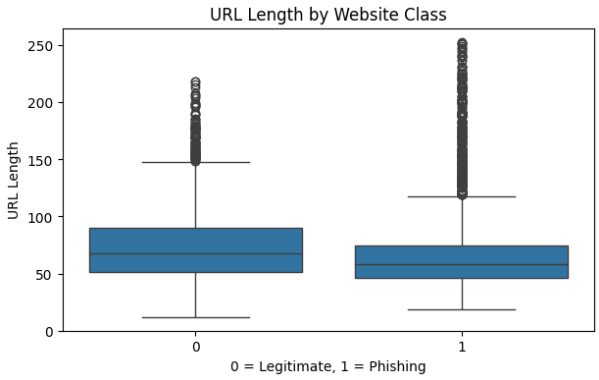


Fig 2. URL length by website class.

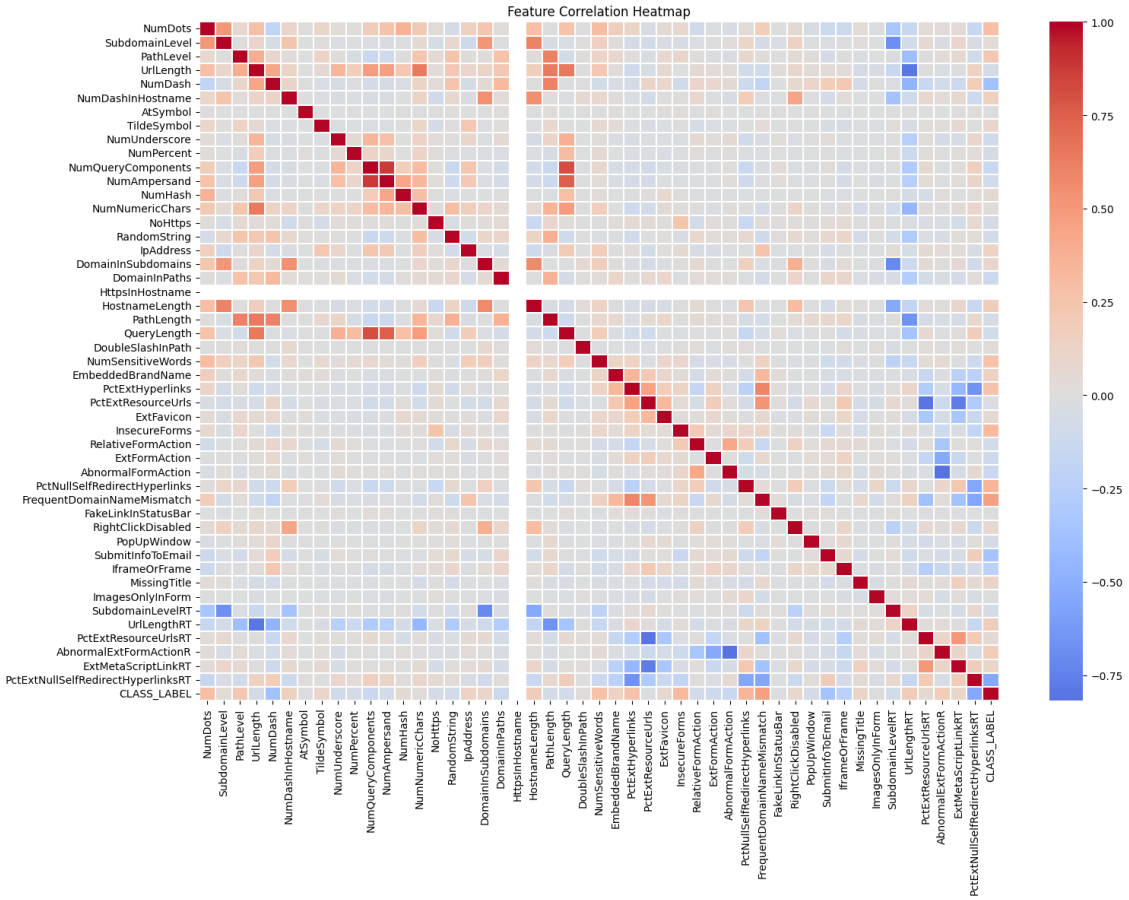


Fig 3. Correlation heatmap.

Key insights

- **URL length alone is weak:** phishing URLs have a lower median (58 vs 68) but a longer tail (max 253 vs 218).
- **Page behaviour matters most:** PctExtNullSelfRedirectHyperlinksRT (-0.54) and FrequentDomainNameMismatch (+0.46) correlate most with the label. Legitimate sites actually use more dashes (NumDash, -0.37).

The 15 selected features

Rank	Feature	Meaning
1	PctExtNullSelfRedirectHyperlinksRT	Rule-coded (-1/0/1) share of external, empty or self-redirect links
2	FrequentDomainNameMismatch	Page domain often differs from the domain in its links (0/1)
3	NumDash	Number of dashes in the URL
4	SubmitInfoToEmail	Form sends the data to an email address (0/1)
5	PctNullSelfRedirectHyperlinks	Share of links that are empty or point to the page itself
6	InsecureForms	Form is submitted over an insecure connection (0/1)
7	NumDots	Number of dots in the URL
8	PctExtHyperlinks	Share of links pointing to other domains
9	NumSensitiveWords	Sensitive words (e.g. login, secure) in the URL
10	IframeOrFrame	Page uses an iframe or frame (0/1)
11	DomainComplexity	New feature: NumDots + SubdomainLevel
12	PathLevel	Depth of the URL path
13	AbnormalExtFormActionR	Rule-coded: form action points to an abnormal external domain
14	UrlLengthRT	Rule-coded (-1/0/1) URL length
15	HostnameLength	Length of the hostname

Table 2. Features kept by SelectKBest (section 4), ranked by ANOVA F-score.

5. Model and Settings Tried

Random Forest (no scaling needed, handles skewed features, gives feature importances). 80/20 stratified split (8,000 train, 2,000 test, random_state = 42). **Baseline:** default settings (100 trees). **Settings tried:** n_estimators = 50, 100, 200, then max_depth = 5 and 10 (both with 100 trees). The top two features, PctExtHyperlinks (0.28) and PctExtNullSelfRedirectHyperlinksRT (0.20), carry about 48% of the importance (100-tree model).

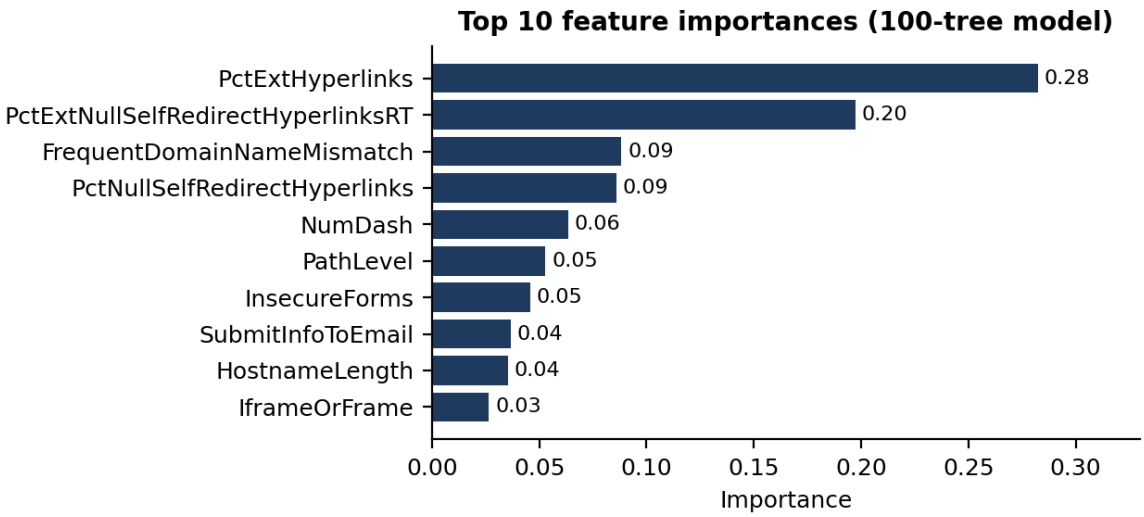


Fig 4. Feature importances (100-tree model).

6. Results

Setting	Accuracy	Precision	Recall	F1	ROC-AUC
n_estimators=200 (final)	0.9775	0.9751	0.9800	0.9776	0.9959
Baseline = n_estimators=100	0.9775	0.9770	0.9780	0.9775	0.9960
n_estimators=50	0.9760	0.9760	0.9760	0.9760	0.9963
max_depth=5 (100 trees)	0.9605	0.9484	0.9740	0.9610	0.9924
max_depth=10 (100 trees)	0.9735	0.9740	0.9730	0.9735	0.9954

Table 3. All settings tried, ranked by phishing Recall (2,000-site test set).

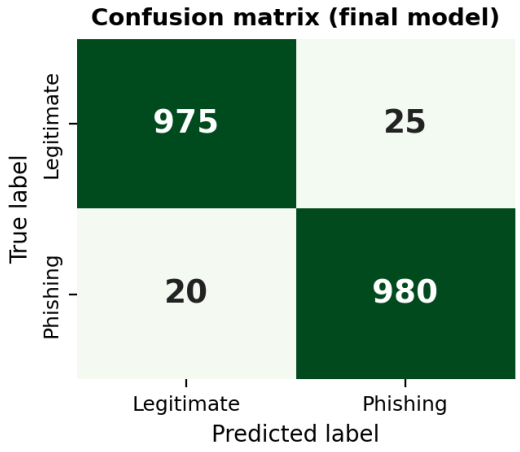


Fig 5. Confusion matrix (final model).

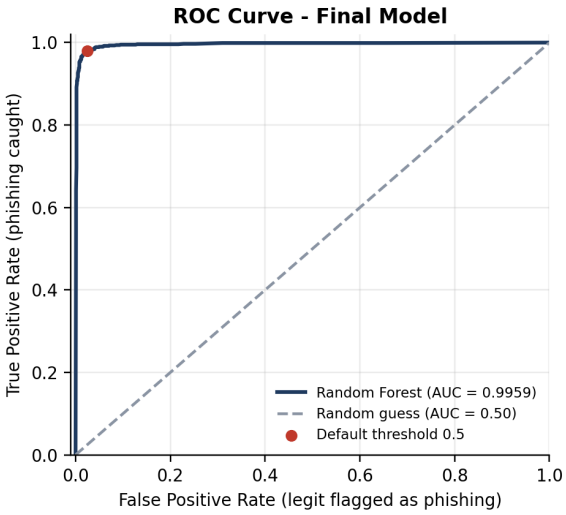


Fig 6. ROC curve (AUC = 0.9959).

- **Confusion matrix and ROC:** 975 TN, 25 FP, 20 FN, 980 TP: only 45 of 2,000 sites were wrong. At the 0.5 threshold the model catches 98.0% of phishing sites and flags 2.5% of legitimate ones.
- **Why 200 trees:** highest Recall (98.0%) with accuracy 97.75%.

7. Conclusion and What I Learned

The final Random Forest (200 trees) reached **97.75% accuracy, 98.0% recall, 97.5% precision and ROC-AUC 0.9959** on unseen websites. What a page does (links to other domains, domain mismatch) is a stronger phishing signal than how its URL looks.

- Tree depth matters far more than the number of trees.
- For security, Recall and the confusion matrix matter more than accuracy alone.
- **Limits:** one split and one algorithm; 419 rows repeat once `id` is removed and may appear in both train and test; features were selected on all rows before the split.